



2. Activate auto-updates

- Navigate to **Settings → Security → Security update** and toggle on “Auto-download over Wi-Fi”.

3. Monitor update receipts

- Install a helper app like **Android Security Bulletin** to receive push notifications whenever Google publishes a new patch.

Example: After purchasing a Pixel 6, Priya confirms her phone is enrolled in the Android Beta program for early monthly updates, ensuring she never misses a critical security patch.

iOS: Streamlined system updates

1. Enable automatic iOS updates

- Go to **Settings → General → Software Update → Automatic Updates**; ensure both “Download iOS Updates” and “Install iOS Updates” are enabled.

2. Use managed update policies (for enterprises)

- Through Apple Business Manager or MDM solutions (e.g., Jamf), enforce supervised devices to install updates within a defined window.

Example: An organization configures corporate iPhones via Jamf to auto-install iOS patches between 2 AM–4 AM to minimize disruption and maintain compliance.

Windows & macOS: Centralized patch deployment

1. Windows Update settings

- Open **Settings → Update & Security → Windows Update**, enable “Automatically download updates, even over metered data connections”.
- For enterprises, deploy the latest Windows 10/11 CIS Benchmark via Group Policy to enforce update deferrals and quality update installation windows.

2. macOS Software Update

- In **System Settings → General → Software Update**,

turn on “Automatically keep my Mac up to date” and click “Advanced...” to select all update types.

3. Leverage patch-management tools

- Use solutions like Microsoft Endpoint Configuration Manager or JAMF Pro to orchestrate and report on patch status across fleets.

Example: A small business uses WSUS (Windows Server Update Services) to schedule Windows updates outside business hours, preventing user interruption while ensuring no critical patches are skipped.

Secure boot and hardware-based trust

Secure Boot and hardware trust anchors prevent unauthorized firmware or OS code from loading during startup. By chaining cryptographic verification from the hardware root of trust up through the bootloader and kernel, you block persistent malware and boot-level compromises.

Android & iOS Secure Boot Chains

■ Android Verified Boot (AVB):

1. Verified Boot v2 performs hash checks on each boot partition image against a public key embedded in hardware.
2. To inspect status, enable **Developer Options → OEM Unlocking** (temporarily), then run `adb shell getprop ro.boot.verifiedbootstate` – it should read “green” for a locked, verified state.

■ iOS Secure Boot:

1. The iPhone’s Boot ROM (immutable in hardware) verifies the Low-Level Bootloader (LLB), which then verifies iBoot and the iOS kernel.
2. Jailbroken devices break this chain, triggering Visible signs (e.g., no OTA updates) and vulnerability to kernel exploits.

Tip: Never enable OEM Unlocking on Android unless you plan to flash a custom ROM; it disables the verified-boot check and opens you to bootloader-based attacks.

